

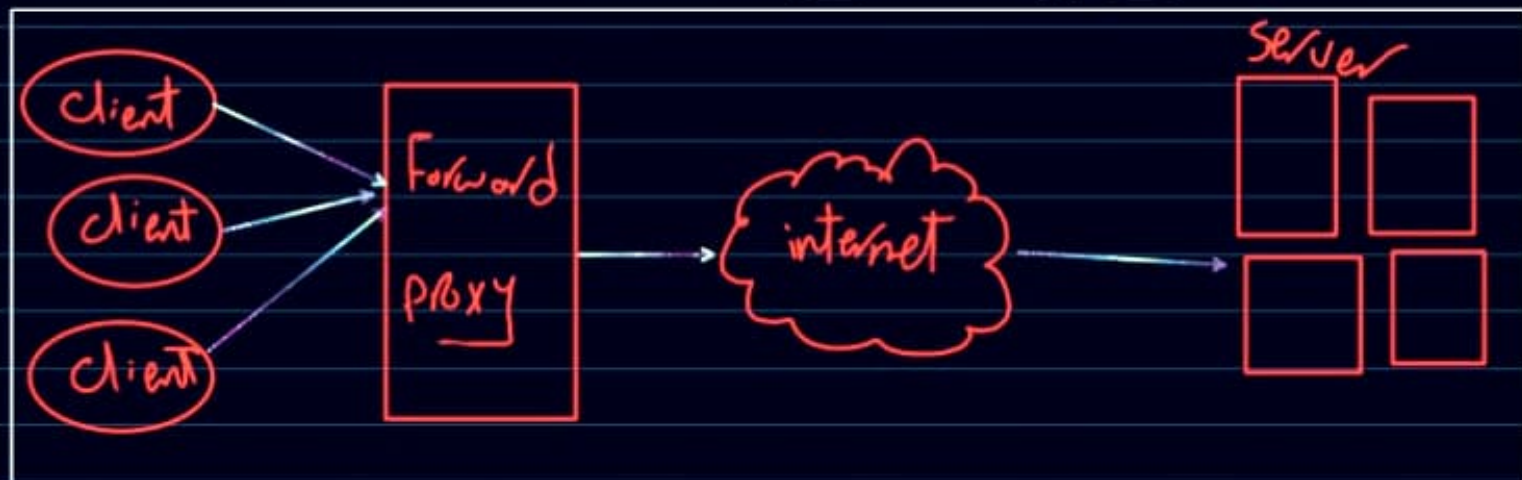
① LTM "F5"

①

✓ Proxy + Load Balancer

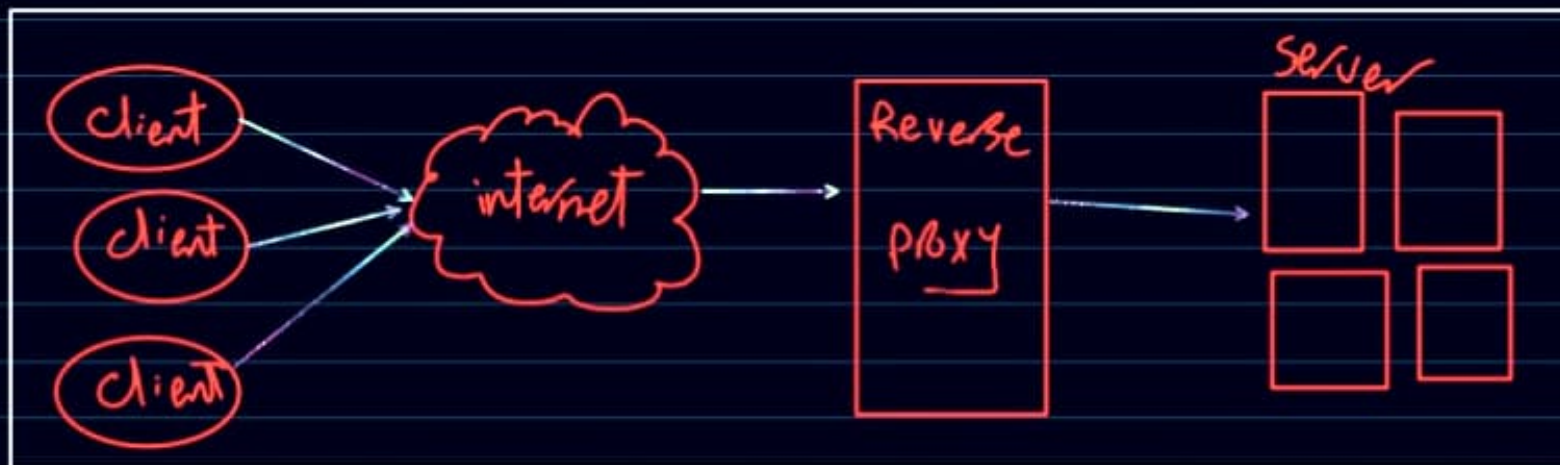
① Forward proxy

- sits between clients and the internet
- used for content filtering, caching and anonymity



② Reverse proxy

- sits between clients and internet servers.
- Protects backend servers and enables load balancing, SSL offload, caching and security
- F5 LTM acts as a Reverse Proxy by default



- Full Proxy

②

client

virtual server

server

① $\xrightarrow{\text{syn}}$
 $\xleftarrow{\text{SYN-Ack}} ②$

③ $\xrightarrow{\text{ACK}}$

④ $\xrightarrow{\text{HTTP-GET}}$

⑤ $\xrightarrow{\text{syn}}$
 $\xleftarrow{\text{SYN-Ack}} ⑥$

⑦ $\xrightarrow{\text{ACK}}$

⑧ $\xrightarrow{\text{HTTP-GET}}$

$\xleftarrow{\text{HTTP-Response}} ⑩$

$\xleftarrow{\text{HTTP-Response}} ⑨$

default IP: 192.168.1.245

(3)

GUI: User: admin password: admin

CLI: User: root password: default

① cli *change MGMTIP *config

② From screen

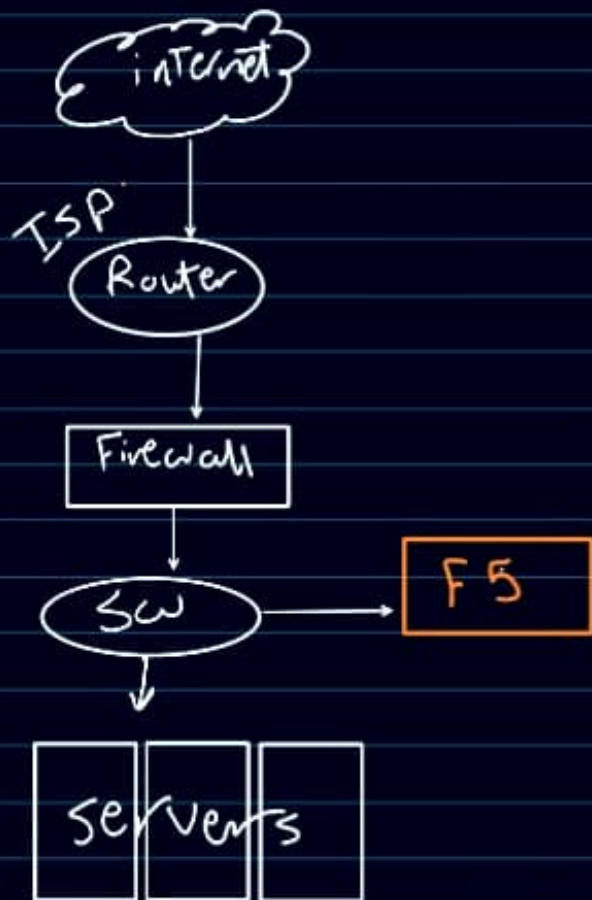
★ license

1. write the registration key
2. Activated Method → Manual
3. Copy the Dossier
4. go To "active.fs.com"
5. paste the Dossier
6. copy the license
7. past the license in FS
8. active

★ Provisioning & modules

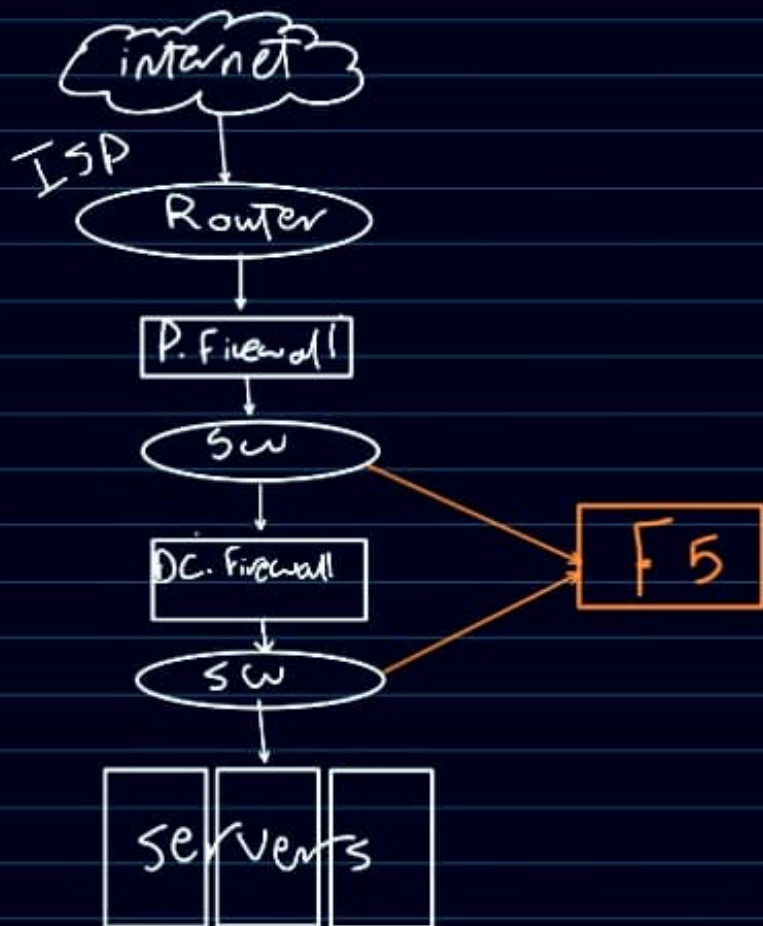
1. choose you Modules "Normal"
2. the will be restarting
3. put the MGMT-IP, Hostname, Root-Password

☆ One Arm Design



☆ Two Arm design

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★ network configuration

- interface → physical interface like: 1.1, 1.2, 1.3
- Trunks → Agg like: LACP
- vlans
- self IP → for Routing, traffic & HA

☆☆ Network tab

1. vlan (name, tag, vlan ID) like ("DMZ-vlan, tagged, 100)
2. self IP assign to vlan
 - port lockdown
 - Allow none no services ✓
 - Allow all any services
 - Allow default default port

★ Steps if I make Agg (trunk)
1. trunk , 2. vlan , 3. self IPs

Load Balance component:

- Node → server IP
- pool member → services (eg. 80, 443)
- pool → Node IP + service
- virtual server → more of pool

⇒ self IP using for Routing in Firewall

★ Adding Route from "network → Route" next hop → Fw

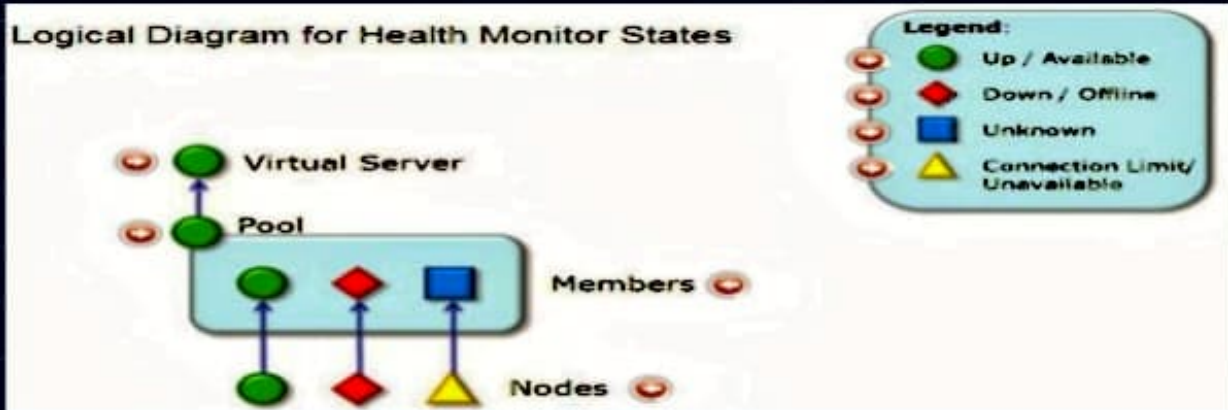
& make a SNAT in BIGIP from virtual server

☆ Health Monitor :

From Pods → health monitor

if I have a custom monitor running by script

• system → File MGMT → External monitor program File → import



• NAT } private → public } snat
private → private }
public → private → Dnat

☆ in LTM → Address translation → SNAT pool list

☆ load balancing

1. Round Robin
2. Fastest
3. last connections
4. observed

• static type → Round Robin, Ratio,

• Dynamic type → Fastest, Last connections, observed

Local traffic → pool → member → LB method

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☆ Priority Group:

1	2	3
①	②	③

• Local traffic → pools → members
& Active "priority Group Activation"

• if I set less than 2 → once 1 or 2, 3 will be active

☆ if I need the Real-IP from user to server

Local traffic → profile → "create custom profile" → parent profile "HTTP"
Enable: insert X-Forward For- + in Request Header Erase "X-Forwarded-For"

☆ Persistence

• Local traffic → profile → persistence → cookie

① Insert: the BIGIP whose insert the cookie

to turn on the persistence "statistics → Module statistics

→ Local traffic → "you show the command copy it and run in the Tmshel:"

☆ type of cookie

Method	who create the cookie	F5's Role	Best use case
① Insert	F5 BIG-IP	injects a new header	standard, default setups.
② Passive	The server	observes existing header	app already manages sessions.
③ Rewrite	The server	overwrites server's header	leggers app compatibility
④ Hash	The server	performs math on header	high-performance, no-touch

☆ one connect: } Connection Reuse overview

Local traffic → profiles → other → one connect

* SSL

👤😊
mmc 8

① first create your certificate

system → certificate Management → create

take a mind in subject Alternative name DNS:www.k.com

} CSR

② after create the certificate by using "Request text" from CA
import it in F5 certificate

③ then attach it in VS

local traffic → profile → SSL → client → create
then add it in VS

iApps :-

What are iApps?

iApps are templates and frameworks within F5's BIG-IP that simplify and automate the deployment and management of application services. They were introduced to move beyond per-device configuration toward an application-centric model.

Key Components

1. iApps Templates (.tmpl files):

- These are essentially scripts (Tcl-based) that guide an administrator through the configuration of a specific application type (e.g., Microsoft Exchange, LAMP stack, or custom web apps).
- They present a simple GUI (in the Configuration Utility) asking for high-level information (like "What is the server IP?" or "What is the domain name?") instead of requiring manual creation of virtual servers, pools, profiles, and iRules.

2. iApps Packages:

- F5 provides pre-built solution templates for various common applications (like VMware View, SAP, or Oracle). These packages contain the template and necessary supporting files.

3. iApps LX:

- This is the next evolution. While classic iApps use Tcl, iApps LX uses Node.js.
- It provides a more modern, RESTful framework for building sophisticated application services and extensions, integrating more deeply with orchestration tools like Ansible, Chef, or cloud platforms.

Why Use Them?

- **Speed & Simplicity:** Deploy complex configurations in minutes rather than hours.
- **Consistency:** Ensures every instance of an application (e.g., a web farm) is configured the same way, reducing human error.

Action F5:

Network → packet filter → default Actions: accept

then in Rules make your Rule

capture on FS :



⇒ `tcpdump -i <interface>`

to list interface

⇒ `tmsh show net interface`

to save the packet capture in file:

⇒ `tcpdump -i <interface> -w /your/path/capture.pcap`

Filter by IP

⇒ `tcpdump -i <interface> host <IP>
src host <IP>
dst host <IP>
port / TCP port`

HA in FS

① create a Vlan in each FS (eg. Vlan 1049 or untagged)

② create self IPs in each FS → allow default

you have 2 selfs (EXT & INT), should make floating for each self
but type of Traffic Group is "Traffic Group (floating)"

③ Devices MGMT → Devices → select your devices → config sync "IP" update
then failover → chose Ext & Int Vlan → update → Mirroring chose HA Vlan

④ in Active Device
Device MGMT → Device trust → Add

